

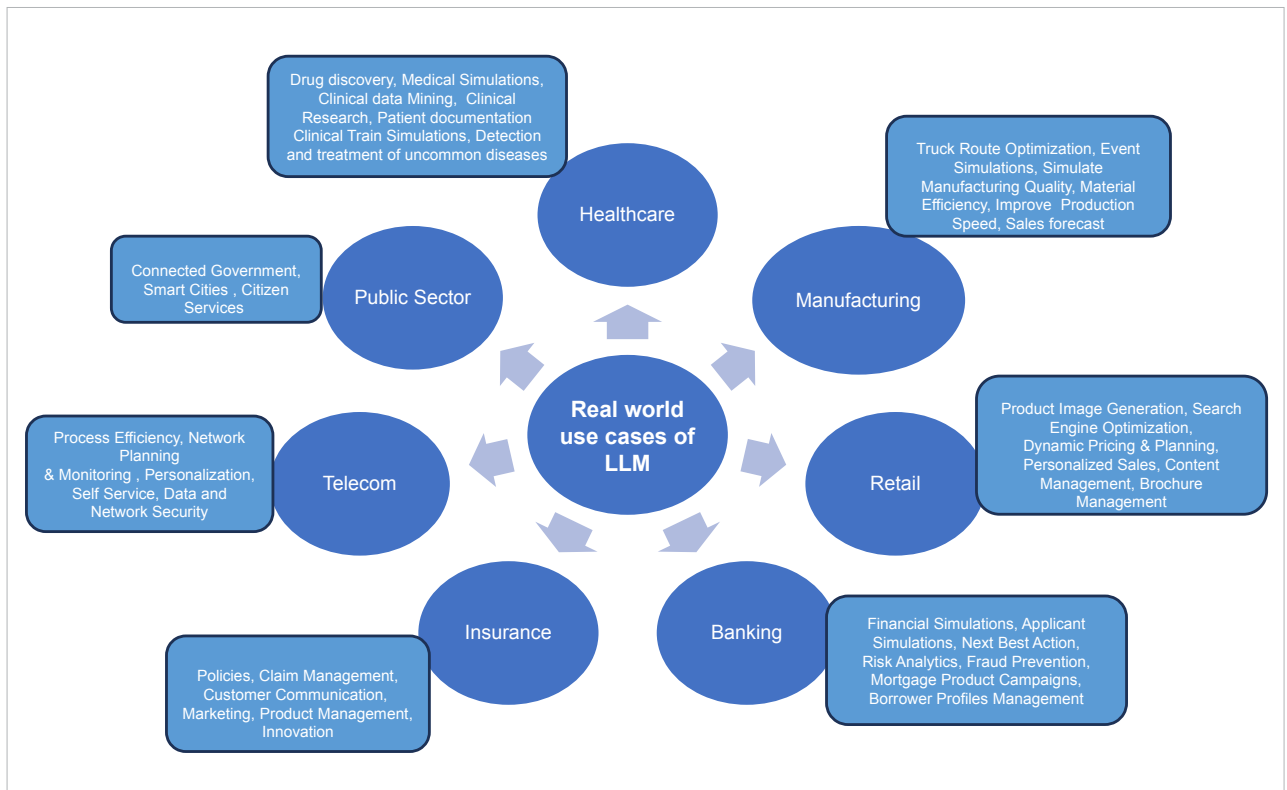


Figure 3: LLM use cases

High cost: Training costs of LLMs range from a couple of million dollars to ten million dollars, making them financially inaccessible for some organisations.

Data quality: Most LLMs have been trained in data based on web scraping. This may lead to plausible but factually incorrect or nonsensical responses, leading to poor data quality.

Data security: Sometimes, LLMs accept publicly available data as input. This can expose enterprise secrets, posing data security risks.

Reliability: Trained models are often 'black boxes', leading to false, harmful, and unsafe results.

Skills gap: Generative AI initiatives requires expertise in machine learning, deep learning, prompt engineering, and large language models, which many enterprises lack in-house.

In summary, LLMs are revolutionising the technology that helps in bridging the gap between

human communications and machine understanding.

The major steps involved in building custom LLM models for an enterprise are:

- Data gathering and preparation
- Training the model
- Evaluating the model
- Integrating with applications
- Deploying the model
- Continuous retraining

The future of LLMs is promising,

powering virtual assistants, improving machine translation quality and enabling innovative applications across various industries. Organisations such as Google, Microsoft, Amazon, Facebook, IBM and OpenAI leverage LLMs for applications like NLP, chatbots, content generation, sentimental analysis, etc. However, LLM technology is very new and not well understood, and many applications remain exploratory. **END** 🐧

Acknowledgements

The author would like to thank Santosh Shinde of BTIS, Enterprise Architecture division of HCL Technologies Ltd, for the support in bringing this article to fruition as part of architecture practice efforts.

By: Dr Gopala Krishna Behara

The author is an enterprise architect in the BTIS Enterprise Architecture division of HCL Technologies Ltd. He has a total of 28 years of experience in the IT industry.

Disclaimer: The views expressed in this article are that of the authors and HCL does not subscribe to the substance, veracity, or truthfulness of the said opinion.